We would like to thank both reviewers for the thorough review and very helpful comments and suggestions. Below we provide point-by-point responses for all comments.

Reviewers' comments:

Reviewer #1: (Comments for the author) This manuscript presents an innovative, accessible tool designed to enhance laboratory safety by automating the identification of chemical hazards with emphasis to reproductive risks, using GHS H-codes and publicly available chemical databases. The manuscript is well organized, written in a clear and professional tone, and describes the workflow in sufficient detail to ensure reproducibility. The integration of PubChem's PUG-REST interface with Python and the tool's applicability across disciplines (neuroscience and chemistry) and hazard types (reproductive, aquatic) are strengths. I also appreciate the attention to hazard interpretation and the discussion of the tool's limitations and intended use in collaboration with EH&S professionals.

I have though some suggestions that would strengthen the manuscript further:

Major points:
1. Benchmark Against Existing Tools: While the motivation for HazardPyMatch is compelling, it would be helpful to briefly compare the tool's features with existing resources (e.g., internal EH&S systems, ChemWatch, or government tools like OSHA or EPA databases). This would better position the novelty and practical value of HazardPyMatch in the broader hazard management ecosystem.

We have updated the Discussion/Limitations section to include two new paragraphs that provide a detailed comparison of HazardPyMatch with existing resources.

2. Quantify Performance and Limitations: Although the manuscript describes some instances of false positives and negatives, I encourage the authors to include a more systematic estimate of performance. For example:
---What percentage of identified hazards were verified as accurate?
---Was any benchmarking dataset used to assess sensitivity or specificity?
Even limited quantitative assessment would reinforce confidence in the tool's reliability.

Thank you for this very valuable suggestion. Toward increased confidence in the tool's reliability, we now include a downloadable sample inventory (Sample_Chemical_Inventory.csv) and ten sample protocols, a benchmarking dataset, on the HazardPyMatch GitHub page that users can utilize to demo the software. The sample inventory is 300 randomly selected chemicals from our master list of chemicals used in neuroscience labs. We quantitatively assess the performance of HazardPyMatch using our benchmarking dataset and provide example output in the HazardPyMatch ReadMe. We introduce the concept of our benchmarking dataset in the first paragraph of the limitations section. While describing the missing CAS Number limitation we provide: "For example, running the benchmarking dataset we provide, users will find that

PubChem is not able to identify the CAS Number for one chemical name, Paraformaldehyde Aqueous Solution". Furthermore, we explain "For example, users who run our benchmarking dataset will find 6 instances of false positives among a total of 28 hazard words found in the 10 provided protocol PDFs" when we discuss the presence of false positives in HazardPyMatch output.

~~3. STAR Methods Compliance: Ensure that methodological sections are explicitly labeled according to the STAR Methods format (e.g., Resource Availability, Method Details, Quantification and Statistical Analysis, etc.) to match journal expectations.~~

Minor points:

4. Persistent Code Access: The .ipynb file provided is appropriate for reproducibility; however, I recommend depositing the code in a permanent public repository such as GitHub or Zenodo with a DOI to improve visibility, citability, and long-term accessibility.

Toward increased access to our tool and consistent with our desire for complete open access, we have deposited the code on github (https://github.com/emp2163/HazardPyMatch). We will set the code accessibility to "public" on the day that the article is published.

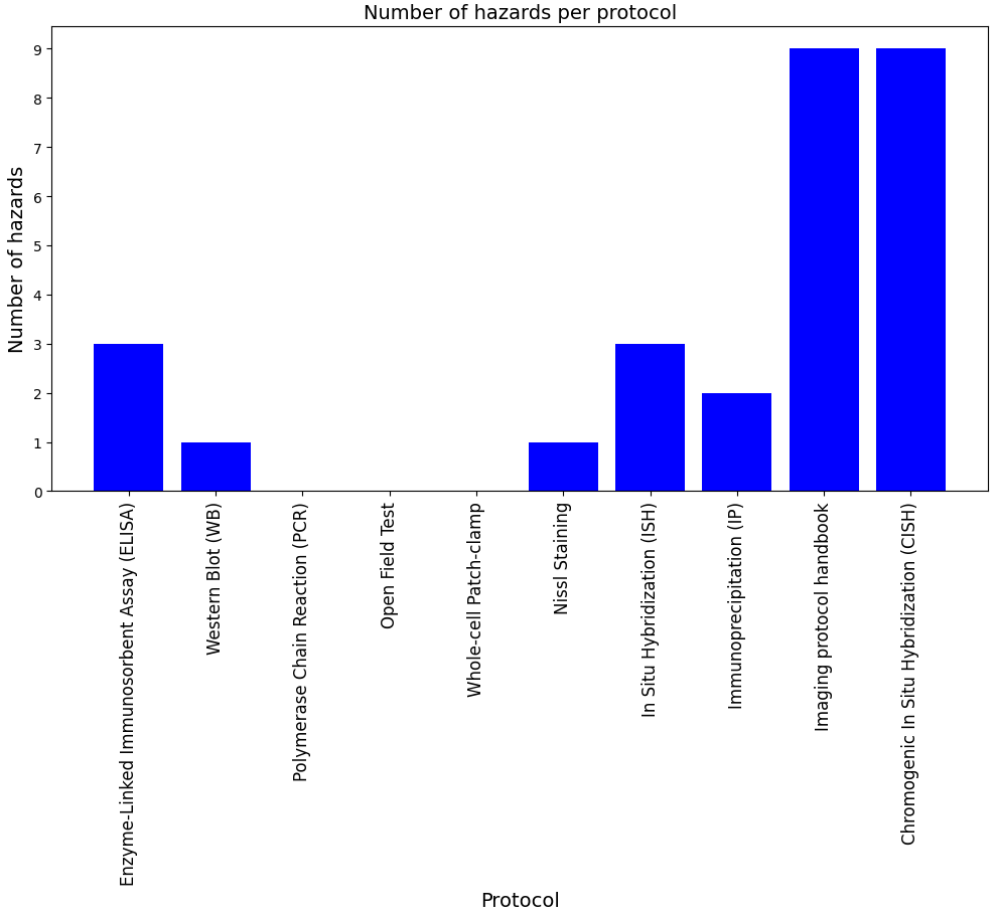
5. Clarify Role of Contextual Risk Assessment: The limitations section is thorough, but it may help to clarify how users can practically integrate contextual information (e.g., dose, exposure frequency, protective equipment use) after hazard identification, or whether this could be a future feature of the tool.

We have now updated the limitations section as follows: "Users can manually consider contextual factors like chemical dose, exposure and frequency based on existing literature; however, they are strongly encouraged to consult their institutional EH&S professionals to ensure a comprehensive, context-specific risk assessment. Future versions of the tool aim to incorporate interfaces for capturing this contextual information directly"

6. Figures and Tables: Consider providing a downloadable version of the bar chart and Pareto chart with embedded figure legends (e.g., for Figure 1), especially if this tool will be widely adopted by labs for protocol-specific risk communication.

Thank you very much for this helpful suggestion. We have now added x-axis labels to the bar and pareto charts — protocol names for the bar chart output and CAS numbers for the Pareto chart output. To maintain readability, our code includes a conditional statement that omits x-axis labels when the number of protocols or CAS numbers exceeds twenty, thereby avoiding overly crowded graphs. By default, we provide a secondary list corresponding to the plotted data, enabling users with larger inventories to identify each bar. We did not include x-axis labels in Figures 1B and 1C of the revised manuscript, as the large number of labels would detract from figure clarity. However, we include on the next page a Figure A - example bar and Pareto charts generated from a

sm ser analyzes a
sm tory.



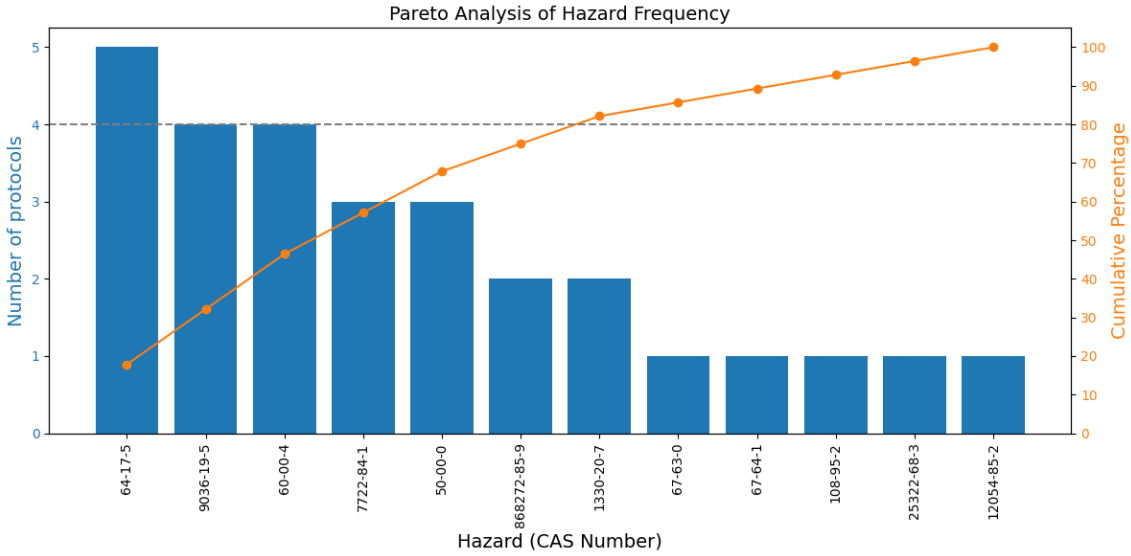


Figure A. Top: Bar chart showing the number of chemical hazards associated with each protocol. The distribution of the data generated from our benchmarking dataset highlights variations in hazard density with a range of 0-9 hazard words found in ten provided protocol PDFs. **Bottom:** Pareto chart displaying the frequency of individual chemical hazards (blue bars) along with the cumulative percentage of occurrences (orange line) indicating a small subset of hazards account for the majority of the occurrences in our benchmark dataset.

Reviewer #2: Review of CR-METHODS-D-25-00092 "HazardPyMatch: A Tool for identifying reproductive and other hazards in scientific laboratories"

In this Commentary, Parker et al. present a screening tool that was developed to enable scientists to identify potential chemical exposures within a laboratory, allowing for a targeted assessment of risk for health effects of concern. While the mandate of Environmental Health and Safety departments in the University setting is to provide education and regulate research conduct to protect workers, including vulnerable populations, these populations may remain unidentified in the work environment because sensitive health information such as pregnancy is kept private. Other individuals may want to take increased precautions because of sensitivities such as a history of cancer. The effect of this on the workforce is likely under-studied (citation would be great if it exists!) but I agree with the authors that assessing risk is critical for fostering safe and inclusive research environments. Anecdotally, an inability to assess risk from a laboratory research environment (or concern about safety) has resulted in some women choosing to abandon bench work completely when planning to start a family. This commentary is well-written, addresses a topic worthy of discussion, and provides a freely available tool that scientists can use to begin to assess risk in their lab environments. Users may encounter several barriers to successful implementation, however, and these are outlined with suggestions in specific comments below.

We thank the reviewer for this thoughtful input. Indeed, the risks of working in a laboratory environment are not well described, and we hope to provide pregnant workers with a practical tool that can be used in a more holistic risk assessment. Such a risk assessment is discussed by Lane et al (Chemical research in toxicology, 2022), and we point the reader to this resource. We now also use this citation to underscore the fact that this is understudied but very important when it comes to closing the leaks in the pipeline. As this reference points out, women are more likely to leave their career in research at a point in their career when they are more likely to expand their families. This could be partly due to the correct assessment of the increased risks of working in a laboratory environment, with risks both to the embryo or fetus and the pregnant worker. Avoiding toxins that may cause harm to the pregnancy would in this case be the correct decision. We suspect that there is also a contribution of the perceived risk of working with substances that may not turn out to cause harm. In this case, continuing to work in the laboratory setting would be the correct decision, whereas avoiding the laboratory environment may cause harm to the progress of one's career. In summary, the correct assessment of risks and being fully informed can be beneficial for both a healthy pregnancy as well as career progress for a group that is particularly vulnerable to slipping through the cracks of the leaky pipeline. The cited resource gives a broader perspective of the situation, and our manuscript describes the practical tool that people need to complete their individual risk assessments.

Methodology & Results

1. Line 7 "We suggest uploading your chemical inventory to Google Drive and using the provided python code with Google's IDE Colab, or similar."

Elaborate on this for those unfamiliar to running python code, to something like "We suggest uploading the correctly formatted Chemical inventory along with the provided python code to

Google Drive, where the code file can be opened and run with Google's IDE Colab, or similar." Also please clarify that the ReadMe with detailed instruction is within the code file.

We appreciate this very helpful comment. In response, we have updated our GitHub repository to include a revised README file as well as a sample inventory file (*Sample_Chemical_Inventory.csv*). The revised README now provides step-by-step instructions with screenshots to guide users through running the code. In addition, we have updated the manuscript to include the sentence: "We suggest uploading the correctly formatted Chemical inventory along with the provided python code to Google Drive, where the code file can be opened and run with Google's IDE Colab, or similar platform" to further clarify this process for readers.

2. Correct formatting of the chemical inventory is essential, and more detailed instructions need to be provided. Clarify that a table of chemical names and associated CAS numbers needs to be obtained- this may be trickier for users at institutions without ChemTracker. It appears to be important that only a single CAS number with no additional symbols is provided in the CAS number column. The chemical inventory from my institution puts % and multiple CAS numbers for products containing mixtures (such as Trizol) in this column, which causes errors.

Thank you for this valuable suggestion. We have revised the README to improve clarity on inventory formatting, now including screenshots that illustrate how to run HazardPyMatch as well as demo outputs that can be reproduced using Sample Chemical_Inventory.csv, ten provided protocol PDF documents, and HazardPyMatch.ipynb.

The code has also been updated to successfully handle multiple CAS numbers (e.g., 50-00-0, 30525-89-4). Symbols listed in chemical inventory input files are processed correctly, with the exception of cases where a chemical name contains a "%" symbol but does not have an associated CAS number. Identifying missing CAS numbers in such cases is beyond the scope of PubChem and PUG-REST. These entries—along with certain names such as Trizol (when no CAS number is included)—are directed to the proprietary reagents list in the first step of the pipeline. We have acknowledged this as a limitation of our tool in the manuscript, stating: "While this represents a limitation of our tool, a conscientious user can identify any missing CAS numbers from sources like Material Safety Data Sheets (MSDSs) and revise the input chemical inventory accordingly. Once updated, HazardPyMatch will successfully process the chemical names and CAS numbers onward to the GHS H-code scraping step."

To complement the new feature for handling multiple CAS Numbers, we added a similar functionality to include information from all possible PubChem compound IDs (CIDs) given there can be more than one CID per CAS Number. Search of the PubChem substance database is also available using our tool and we pull PubChem substance IDs (SIDs) for thoroughness. Although, note that SID entries can be very numerous, typically are very short and often do not include GHS information.

3. Because formatting of the Chemical Inventory is essential, please provide an example in supplemental data that can be used for trial run of software.

Please see responses to comments 1 and 2, above.

4. "HazPyMatch will load the chemical inventory and fill in any missing CAS Numbers". In my hands the step where it searched for CAS numbers took a long time if they were missing, but they were not successfully filled in. Either code needs to be fixed, or more instruction provided on formatting of chemical names so this works.

We appreciate the reviewer's thoughtful observation and guidance. To improve performance, we implemented parallel processing to accelerate computation. However, some time limits remain necessary due to request throttling by the PubChem API. To support users during longer processing steps, we include progress logging. Since this tool is designed for individual scientists working with a single laboratory inventory, we believe that the optimizations described above sufficiently address performance needs, and additional features are not required.

To address the reviewer's experience of unsuccessful CAS number retrieval, we replaced the Python dependency previously used for this task with a more robust method that directly queries PubChem for missing CAS numbers.

5. Starting at "Our first HazardPy Match application was to identify reproductive hazards…" this paragraph is no longer describing obtaining and uploading the chemical inventory but rather describing an application of the tool. Consider breaking the parts about running the tool and providing GHS codes into new paragraph. The final part of the paragraph describes the chemical inventory provided, but is unclear which merging was conducted to create the inventory (which was provided to the tool), and which steps were performed by the HazPyMatch tool. Please re-word.

We have restructured and reworded this section for improved clarity. Specifically, we moved the paragraph beginning with "Our first HazardPyMatch application was to identify reproductive hazards…" to a later section, while retaining the description of the chemical inventory within the *Obtaining and Uploading a Chemical Inventory* section. Additionally, we replaced the term "merged" with "included," as no merging occurred in the code. The chemical inventory was provided by EH&S and included chemicals from all laboratories within our institute.

6. In "Standardizing Chemical Nomenclature and Handling Redundancy" it is not clear what is meant by "a redundant information handling step", and where the chemical name synonyms are compiled in the inventory. How are chemicals with same name, different CAS numbers handled? For example, free EDTA (60-00-4) has a different CAS number than EDTA trisodium salt (150-38-9) with the former associated with reproductive hazard and the latter not, but both would have the name EDTA.

We have removed text mentioning redundancy as, upon our reading of this again, and considering the reviewers comment, this text introduces uncertainty and distracts from the description of the tool. Additionally, we have taken the liberty to resolve a similar issue to the one the reviewer describes - the "Standard Chemical Nomenclature and Handling Redundancy" header and other descriptions in the Figure 1A are too dense and

could confuse the reader. We therefore substantially simplified Figure 1A so it allows the reader to more quickly and efficiently identify the salient features of the tool.

To address the reviewer's question about multiple Chemical Names with the same CAS number, the previous version of the code actually did this, provided there is only one chemical name per row in the input list. As it is possible that chemical inventories may have more than one Chemical Name per row, we highlight that HazardPyMatch requires one Chemical Name per row in Figure 1A and in the HazardPyMatch ReadMe document included on GitHub.

7. Please provide an example of "Output 1" - could be supplemental. The inclusion of only Table 2 is confusing, because this is not a direct output from the tool.

We have significantly updated the README file provided with the code, which now includes an example chemical inventory and screenshots of the outputs (including Output 1) to help users complete all steps of the pipeline. An example of the first ten entries in Output 1 is also provided below in Figure B. We have clarified in the text that Table 2 is not a direct output of the code, but rather the result of our manual literature search. The revised text reads: "Table 2 integrates data from Output 2 of HazardPyMatch with additional information gathered through manual literature review, detailing each chemical's adverse effects, exposure, timing, and safety considerations."

Chemical Name	CAS Number	PubChem ID	H Codes	H_codes_found	has_relevant	matched_relevant
Ethanol	64-17-5	702	H335: May c	['H225', 'H315', 'H319',	TRUE	['H340', 'H360']
Formaldehyde	50-00-0	712	H350: May c	['H220', 'H227', 'H280',	TRUE	['H341']
Methanol	67-56-1	887	H311: Toxic i	['H225', 'H301', 'H302',	TRUE	['H360']
Phenol	108-95-2	996	H311: Toxic i	['H301', 'H302', 'H311',	TRUE	['H340', 'H341', 'H360']
Phenol	108-95-2	7966	H316: Cause	['H302', 'H303', 'H312',	TRUE	['H361']
Paraformaldehyde	30525-89-4	712	H350: May c	['H220', 'H227', 'H280',	TRUE	['H341']
Ethylenediaminet etraacetic acid (EDTA)	60-00-4	6049	H335 (31.8%	['H302', 'H303', 'H312',	TRUE	['H361']
Isopropyl alcohol	67-63-0	3776	H335: May c	['H225', 'H303', 'H305',	TRUE	['H361']
Acetone	67-64-1	180	H335: May c	['H225', 'H305', 'H319',	TRUE	['H361']
Triton X	9036-19-5	24775	H314 (100%	['H302', 'H314', 'H361',	TRUE	['H361']

OUTPUT 1 Preview

Figure B. Example Output 1 from HazardPyMatch. Output 1 is a Chemical Inventory containing only user chosen GHS H-Codes. Here we are displaying the first 10 out of 136 total chemical names identified using the benchmarking dataset.

8. In "Mapping Chemicals to Experimental Protocols" more detail is needed about what information needs to match previous steps for a chemical to be identified in the protocol. Is it

chemical names identified in previous steps, and is exact match required? Or is CAS number required? None were identified in the lab protocols I tested.

We appreciate the reviewer's insightful comment and explanation of the errors. To clarify, both chemical names and all associated synonyms are used to search for hazards in the protocol PDFs. To improve clarity, we added the following sentence to the *Mapping Chemicals and Experimental Protocols* paragraph: "During this step, the documents are scanned for chemical names and their associated synonyms." It is likely that the reviewer's earlier difficulties running the code resulted in too small a list to enable identification of chemicals and synonyms within the protocol PDFs.

9. "Extended Applications: Aquatic Life Hazards…"

The extension of this tool's use to the zebrafish model doesn't strike as particularly useful, as lab animals are not generally exposed to most reagents in the lab. Perhaps describe as a way to identify chemicals that should be avoided/used with caution in aquatic facilities.

We have updated the text to clarify that these are chemicals whose use should be avoided or approached with caution in aquatic facilities. Specifically, we added the following sentence: "We evaluated aquatic life hazards to demonstrate how the tool can identify chemicals that should be avoided or used with caution in aquatic facilities."

Limitations of Study & Discussion

This is a nice discussion of limits of the tool, ultimately it provides a solid way to begin thinking about risk in one's research environment, and this is important.